Nonlinear Dynamics & Chaos HW5

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Technion

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1 Rossler System

1.1 Fixed point near the origin

Integrating using the python script the following fixed point was found near the origin:

$$x = 0.007, \qquad y = -0.035, \qquad z = 0.035 \tag{1}$$

whose jacobian eigenavlues are

$$\lambda_{1,2} = 0.1 \pm i, \qquad \lambda_3 = -5.7 \tag{2}$$

with eignevectors

$$v_1 = \begin{pmatrix} 1 \\ -0.1 - i \\ 0.005 - 0.001 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 1 \\ -0.1 + i \\ 0.005 + 0.001 \end{pmatrix}, \qquad v_3 = \begin{pmatrix} 0.17 \\ -0.03 \\ 1 \end{pmatrix} \tag{3}$$

The two first eigenvalues produce a slow unstable spiral on the xy plane since the real part is very small but positive, and the eigenvector is mostly in the xy plane, and the third eigenvalue produces a fast strongly stable direction in z since it is strongly negative compared to the other eigenvalues and the corresponding eigenvector is strongest in the z direction.

1.2 Varying c

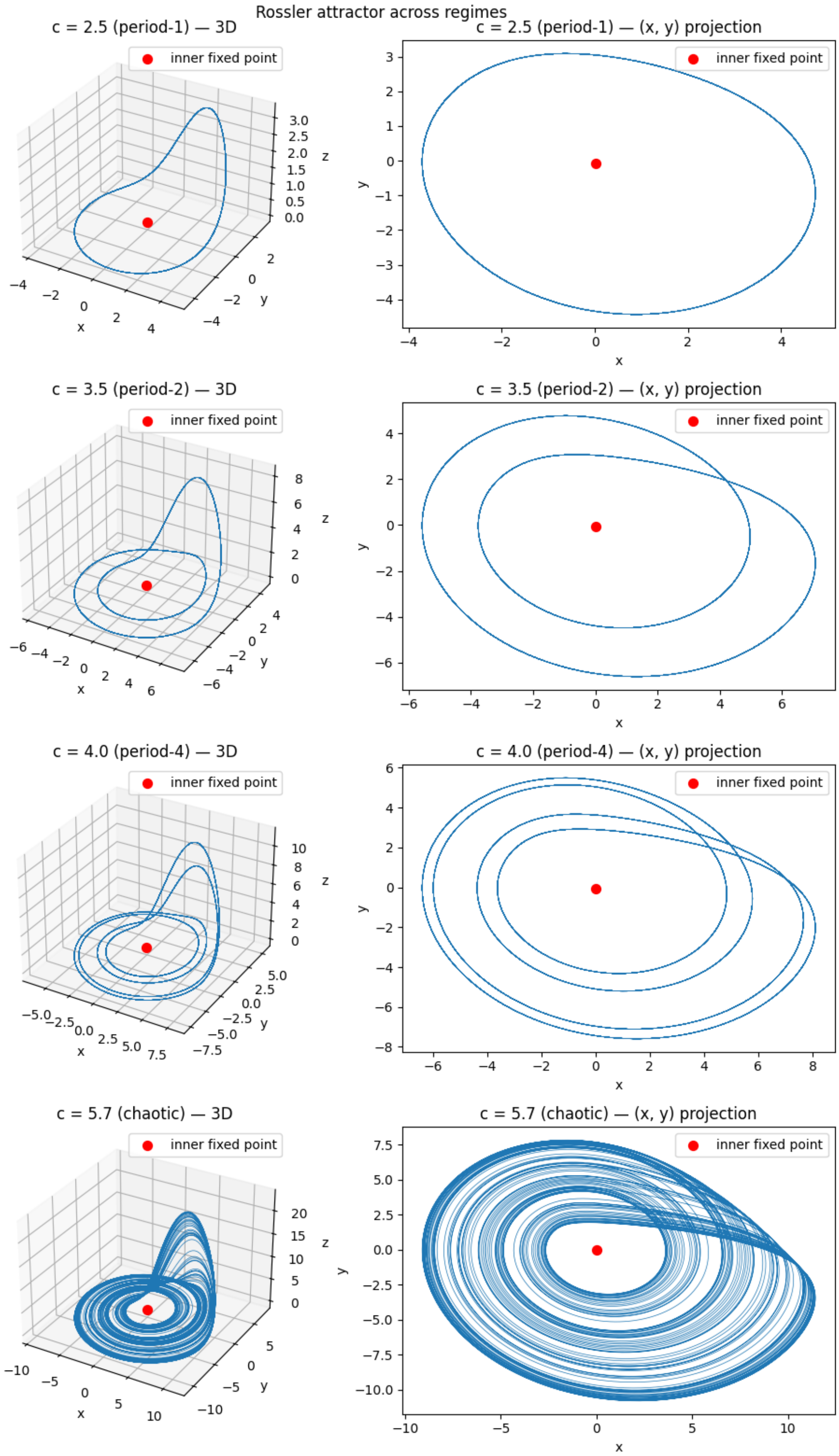


Figure 1 – Trajectories for different c values in 3D and on the xy plane.

1.3 The chaotic attractor

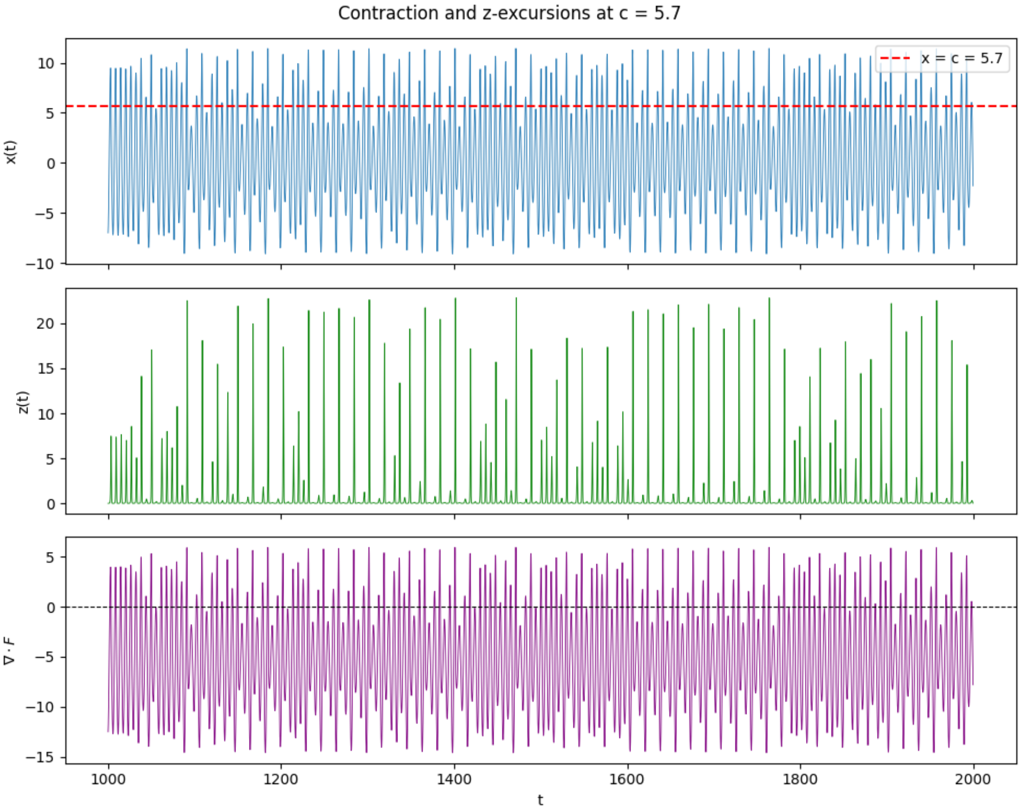


Figure 2 – x, z, and the divergence plotted over time. A red dashed line for x=c is presented and a black dashed line for div=0 is presented.

1.3.1 Average contraction

Calculating the average, minimum, and maximum of the divergence I found

$$\nabla \cdot F_{avg} = -5.3, \qquad \nabla \cdot F_{\min} = -14.6, \qquad \nabla \cdot F_{\max} = 5.9 \tag{4}$$

So while the divergence is on average negative, it is not negative everywhere.

1.3.2 The excursions from the $z = 0$ plane

after plotting $x(t)$ and $z(t)$ I saw that there seems to be correlation between when x spikes and when z spikes. Looking at the equations, it's clear $\dot{z}$ increases when $x > c$.

1.4 Lyapunov exponents and dimension

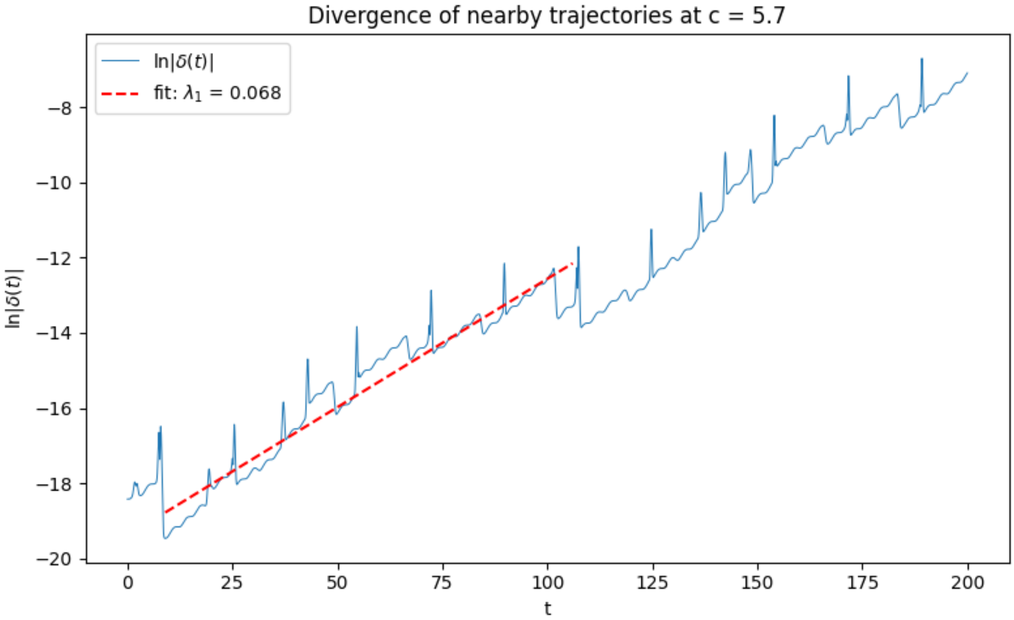


Figure 3 – $\ln|\delta(t)|$ plotted over time and a fit for λ_1.

From the fit we find $\lambda_1 = 0.068$, and we were given $\lambda_2 = 0$. From this initial slope we can estimate the prediction horizon as

$$T_{pred} = \frac{1}{\lambda_1} \cdot \ln\left(\delta_{\max}/\delta_0\right) = 269.6 \tag{5}$$

then we can find *lambda*$_3$:

$$\lambda_3 = \langle \boldsymbol{\nabla} \cdot \mathbf{F} \rangle - \lambda_1 - \lambda_2 = -5.313 - 0.067 - 0.0 = -5.38 \tag{6}$$

and therefore the kaplan-yorke dimension was calcualted to be

$$D = 2.0125 \tag{7}$$

1.5 Poincare section

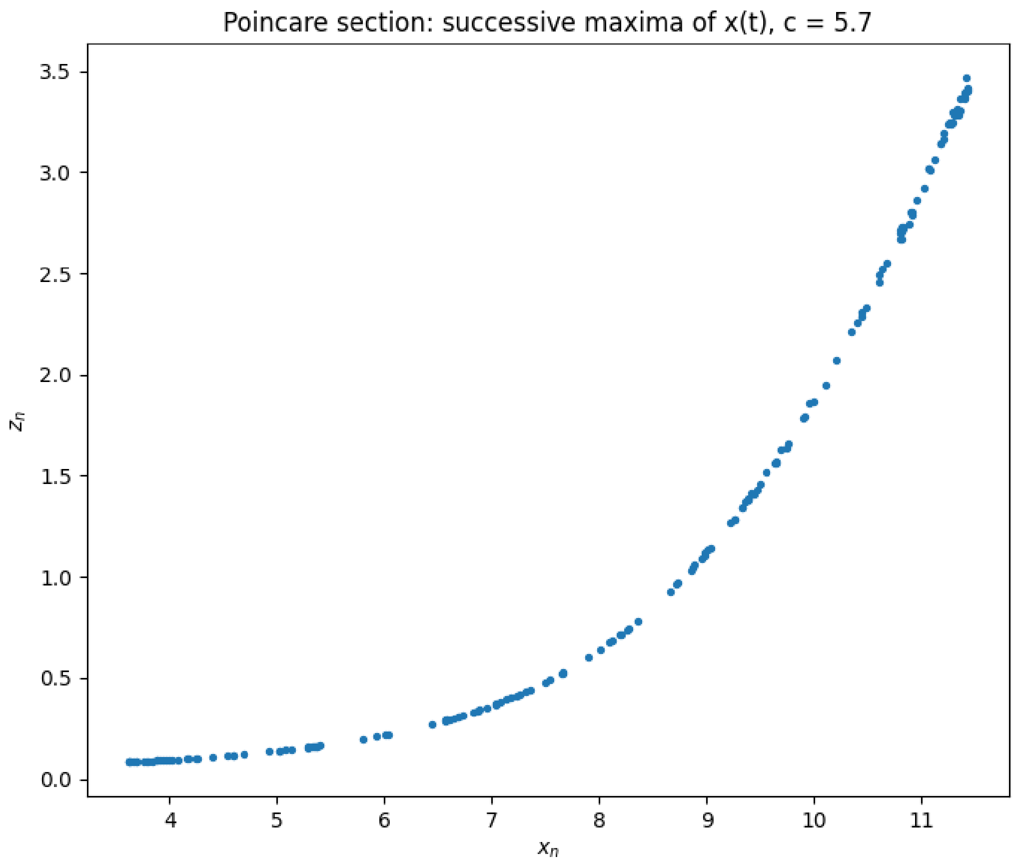
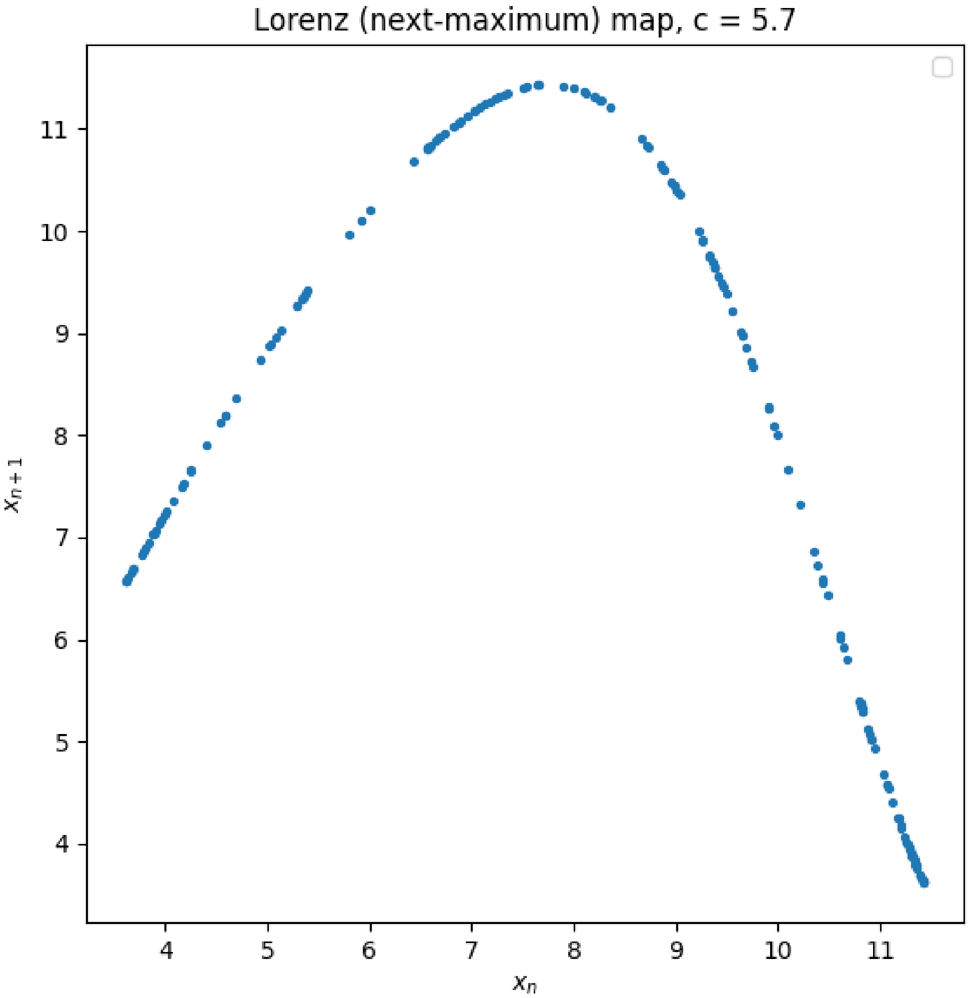


Figure 4 – successive maxima of x plotted in the xz plane. This fits what I expected to see, as the "holes" in the plot imply the spiky structure I saw in part c. In region beyond $x \sim c$ we see z_n rise sharply.

1.6 Lorenz map



which quite clearly satisfies the requirement to be a nearly one-dimensional single-humped map.